Ethan Tapia

657-256-2947 | Anaheim, CA | etapia@chapman.edu | Portfolio | LinkedIn | GitHub

Education

Chapman University — Orange, CA
Bachelor of Science in Computer Science, Themed Inquiry in Design and Fabrication — *May 2026*

Technical Skills

Languages: C#, C++, C, Python, Java, MATLAB, R, SQL
Industrial Controls: Beckhoff TwinCAT, EtherCAT, PLC Programming, Elmo Servo Drives, Embedded Control Systems, H.I.L. Testbed Development
Frameworks & Web: .NET, Next.js
Libraries: OpenCV, DLTdv, NumPy, pandas, Matplotlib
Developer Tools: Git, GitLab, Bitbucket, Bamboo, Jira, Confluence, Docker, Splunk, VSCode, MATLAB Simulink, AutoCAD, SolidWorks, Vectorworks, xLights
Practices: System Testing, Regression Testing, Acceptance Testing (ATP), Root Cause Analysis, Real-time Systems, Digital Logic Design, State Machine Design

Professional Experience

Software Engineering Intern — June 2026 – Present
The Walt Disney Company — *Orlando, FL*

- Owned roughly 60% of tickets for the first production release (Nov. 2026) of a real-time attraction control system, applying object-oriented design in C#/.NET and validating on Beckhoff TwinCAT/EtherCAT
- Developed C++ features applying object-oriented design to real-time animatronic control systems, validated on Beckhoff environments interfaced with Elmo servo drives
- Owned the full TwinCAT rewrite of an attraction control system: design assurance (DA) kickoff, system design reviews, code rewrite, theory-of-operations documentation, and ATP execution per company standards
- Migrated a decade-old ride-control diagnostics tool integrated with Maximo logging onto a React/Node.js/MongoDB stack, closing a technical-debt ticket open for nearly a year and improving fault-identification tooling
- Partnered with Walt Disney Imagineering (WDI) to test and commission control, software, and animatronic systems on live and in-development attractions, authoring and executing ATPs and diagnosing software and hardware issues

Systems & Software Engineering Intern — June 2025 – Oct. 2025
Diality Inc. — *Irvine, CA*

- Performed system-level testing and built hardware-in-the-loop testbeds for hemodialysis device validation, ensuring compliance and subsystem integration
- Conducted root cause analysis of system failures using Python, C++, and cloud log data to improve robustness
- Developed C++ test automation tools that improved testing efficiency by 30% across legacy and next-generation systems

Engineering Research Assistant — July 2023 – Feb. 2025
Chapman University — *Orange, CA*

- Researched fish-scale patterns for prosthetic applications: designed scales in CAD, ran tensile testing on various scale patterns, and used computer vision to analyze hydrodynamic efficiency
- Built engineering prototypes and real-time control simulations using C++, MATLAB Simulink, Python, and R
- Applied advanced testing and data-driven feedback loops to evaluate performance, securing a $4,000 research grant

Property Manager — June 2020 – March 2026
Fairgrove Property Management — *Irvine, CA*

- Coordinated technicians, contractors, and vendors on improvement projects; modernized operations with AppFolio, boosting productivity 20% while sustaining 98% occupancy for 5 years

Projects

Heritage Fest | *C++, Python, AutoCAD, Embedded Controllers*

- Led development of a large-scale projection-mapped light show; programmed embedded controllers with state machines for real-time audiovisual synchronization
- Designed and fabricated weatherproof enclosures housing DMX512 control hardware for reliable outdoor, multi-season operation

F.I.S.H.Bot | *C++, Python, MATLAB, Simulink, R, AutoCAD*

- Developed an autonomous swimming robot, implementing motor control and sensor integration in C++ with MATLAB Simulink for real-time simulation and control tuning
- Built particle image velocimetry (PIV) testbeds to analyze water flow and optimize hydrodynamic efficiency through iterative hardware-software integration